Unsaturated hydrocarbons – Reactions of alkenes- Part 1

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Types of Reactions

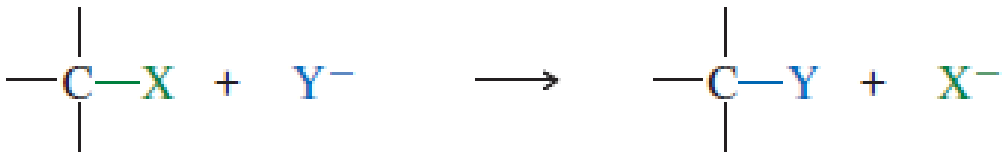
- There are three general types of reactions that organic compounds can undergo:

    i. *Addition reactions*
    ii. *Substitution reactions*
    iii. *Elimination reactions*

Substitution and Elimination Reactions

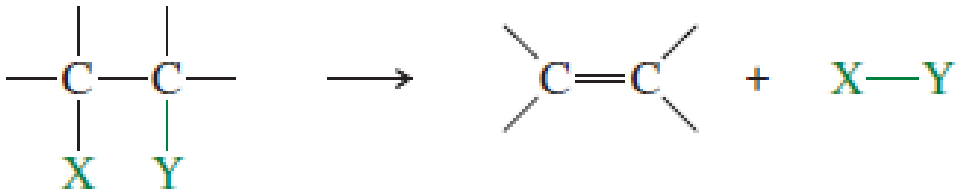
- In a substitution reaction, one atom/group of atoms replaces another atom/group of atoms in a molecule;

Substitution



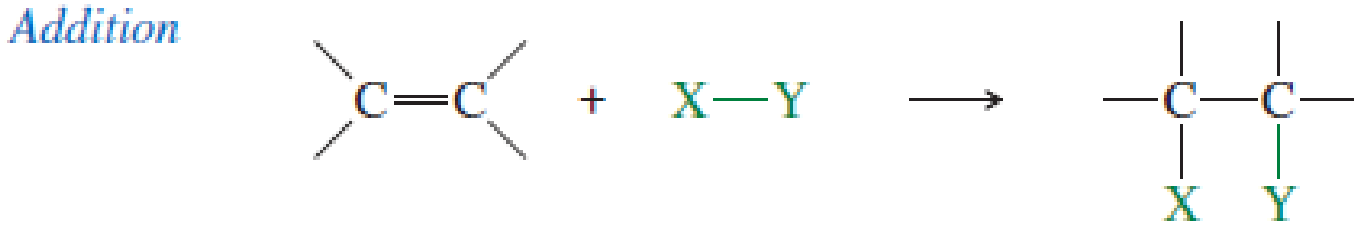
- In an elimination reaction, one molecule splits into two molecules;

Elimination



Addition Reactions

- In an addition reaction, two groups add to the carbons of the double bond and the carbons become saturated.

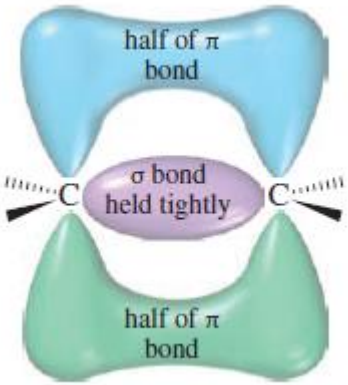


- The pi bond is broken to form ***two new*** sigma bonds to two new atoms/groups of atoms.

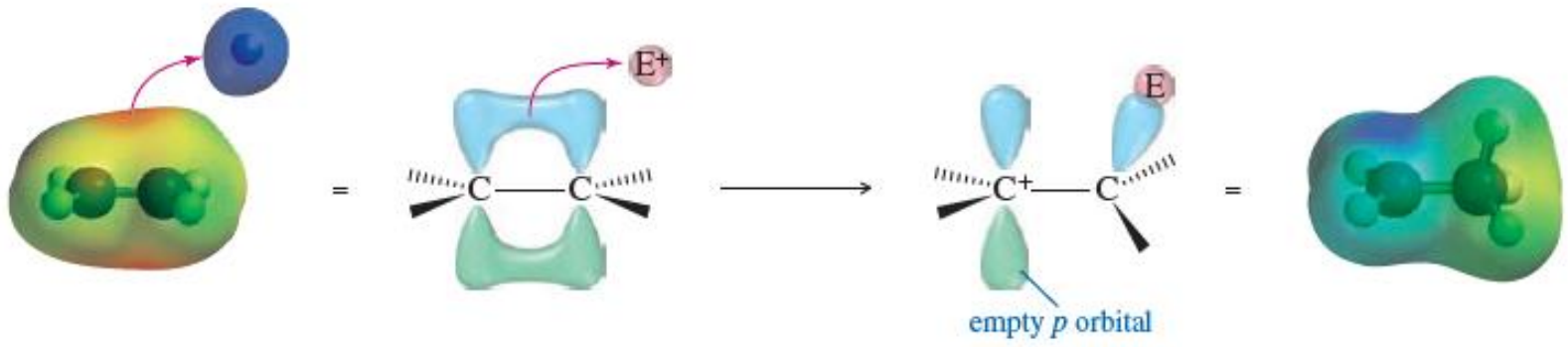
- Addition reactions are the most common reactions of the alkenes.

Addition Reactions of Alkenes

- The presence of double and triple bonds increases the hydrocarbons reactivity.
- Alkenes act as nucleophiles, due to the presence of the pi bond.
- To explain this, consider the structure of the pi bond:
- The pi electrons are delocalised (spread out) above and below the sigma bond. This is unlike in a sigma bond, where the electrons are tightly held.

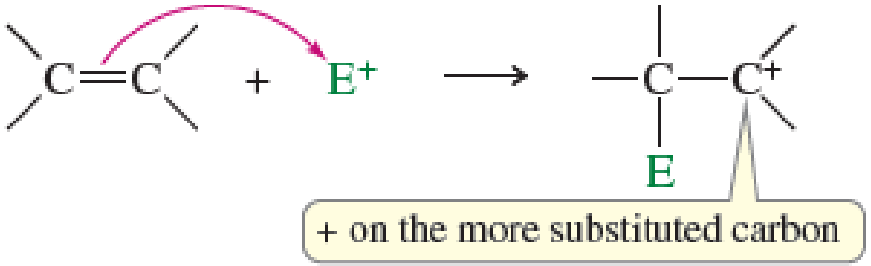


- Therefore, the pi electrons are located further from the carbon nuclei, and are therefore loosely held (and hence easy to donate).
- Electrophiles have an affinity for these loosely held electrons.
- It can pull them away to form a new bond, leaving one of the carbon atoms with three bonds and a positive charge (a carbocation).
- In effect, the double bond has acted as a nucleophile, donating a pair of electrons to an electrophile.
- This type of addition reaction is known as an ***electrophilic addition*** because an electrophile is adding to a nucleophile.

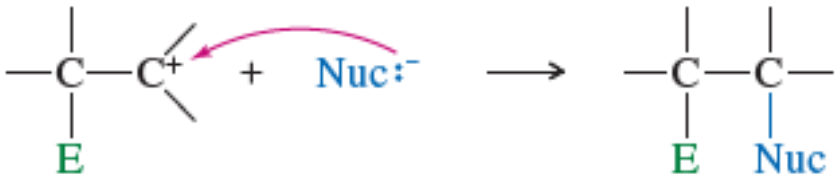


- Most addition reactions involve a second step in which a nucleophile adds to the carbocation to form a stable product.
- The general, key reaction mechanism for the electrophilic addition of electrophiles to alkenes is shown below:

Step 1: Attack of the electrophile on the double bond:



Step 2: Attack by the nucleophile gives the addition product:



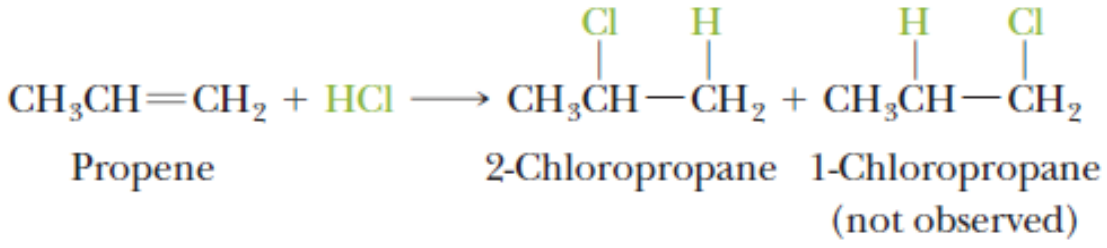
- Several examples of addition reactions of alkenes are shown in the table below, along with the descriptive names associated with each:

Reaction	Descriptive Name(s)
$>C{=}C< + HX \longrightarrow -C(H)-C(Cl\ (X))-$ ($X = Cl, Br, I$)	Hydrochlorination (hydrohalogenation)
$>C{=}C< + H_2O \longrightarrow -C(H)-C(OH)-$	Hydration
$>C{=}C< + X_2 \longrightarrow -C(Br\ (X))-C(Br\ (X))-$ ($X_2 = Cl_2, Br_2$)	Bromination (halogenation)
$>C{=}C< + BH_3 \longrightarrow -C(H)-C(BH_2)-$	Hydroboration
$>C{=}C< + H_2 \longrightarrow -C(H)-C(H)-$	Hydrogenation (reduction)

Addition of hydrogen halides to alkenes

- The hydrogen halides (HCl, HI, HBr) add to alkenes in an *electrophilic addition* to give haloalkanes (alkyl halides).
- For example, ethane reacts with HCl to give chloroethane.

- The addition of HCl to propene gives 2-chloropropane as the major product. 1-chloropropane is hardly observed (very minor product);

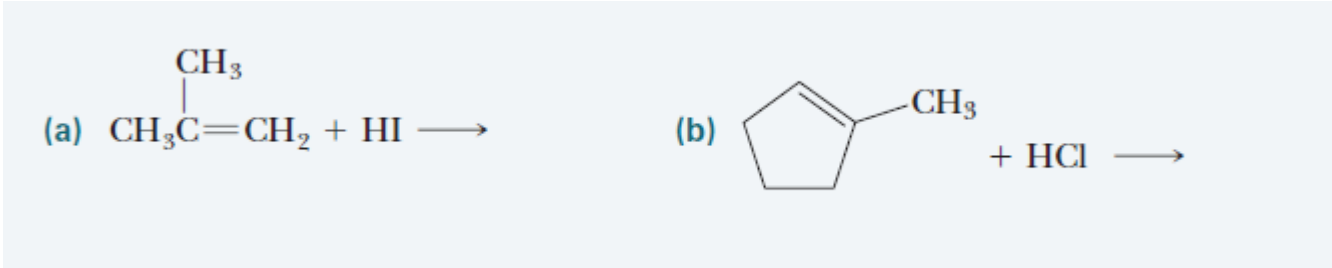


- We say that the addition of HCl to propene is highly ***regioselective***.

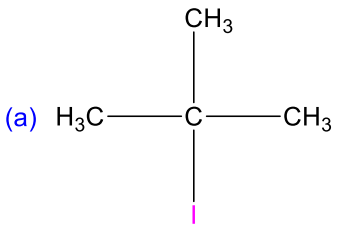
- A *regioselective* reaction is one in which one direction of bond forming and breaking occurs in preference to all other directions. So one of the two possible orientations of addition results preferentially over another.

- Markovnikov's rule: *the generalisation that in the addition of hydrogen halides (HX) to alkenes the* ***hydrogen will add*** *to the carbon of the double bond that has the* ***greater number of hydrogens*** *bonded to it compared to the other carbon of the double bond*.

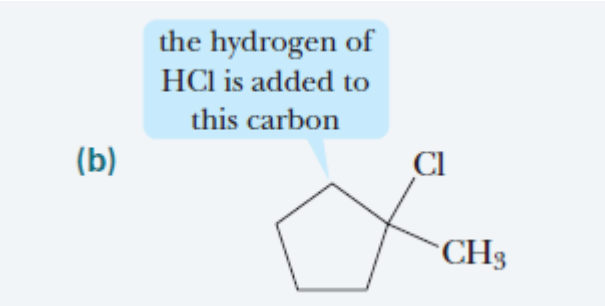
- Practice exercise 1 : Name and draw the structural formula for the major product for each alkene addition reaction:



Ans: a) 2-iodo-2-methylpropane



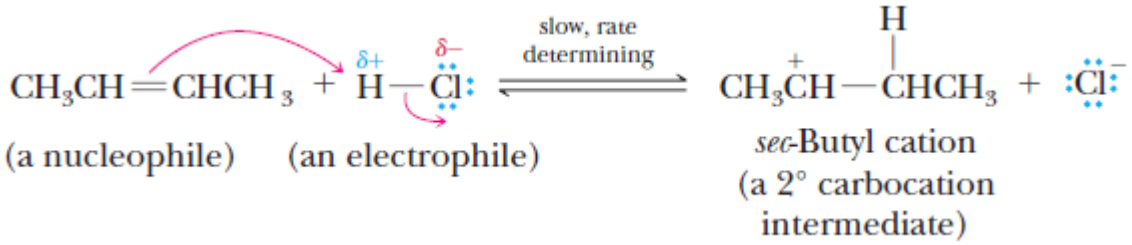
b) 1-chloro-1-methylcyclopropane



- The addition of HX to an alkene proceeds by way of a two-step mechanism:

For example, for the addition of HCl to 2-butene:

Step 1: Add a proton:



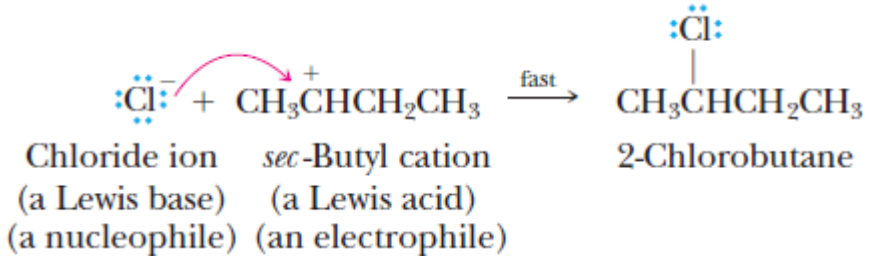
The double bond is a site that is rich in electron density. It acts as a nucleophile.

The double bond (nucleophile) attacks the partially positive hydrogen of electrophilic species HCl. This is shown by the first curved arrow, which shows that the pi bond is breaking, and the electron pair in that bond is used to form a new covalent bond with the hydrogen of HCl.

NB! Keep in mind Markovnikov's rule

The second curved arrow shows the breaking of the polar H-Cl bond, with both electrons in that bond being given entirely to Cl, forming Cl- ion (heterolytic cleavage).

Step 2: Reaction of an electrophile and nucleophile to form a new covalent bond:



The chloride ion uses a lone pair of electrons to form a new covalent bond with the positively charged carbon of the carbocation.

Draw Mechanisms

HOW TO

Mechanisms show how bonds are broken and formed. Although individual atoms may change positions in a reaction, the curved arrows used in a mechanism are only for the purpose of showing electron movement. Therefore, it is important to remember that curved arrow notation always shows the arrow originating from a bond or from an unshared electron pair (not the other way around).

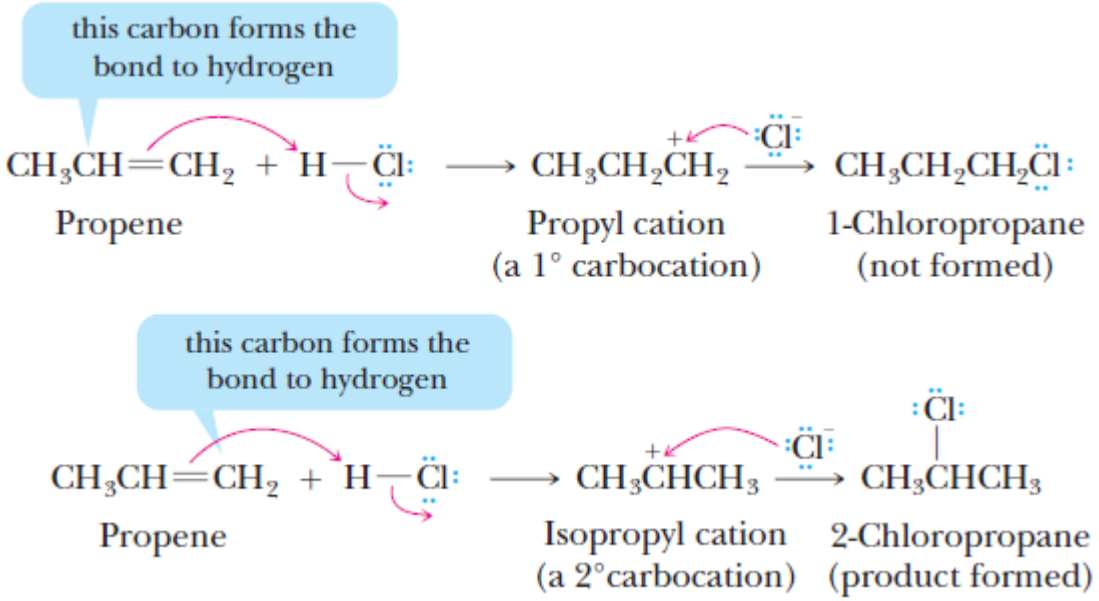
Correct use of curved arrows...

Incorrect use of curved arrows...

a common mistake is to use curved arrows to indicate the movement of atoms rather than electrons

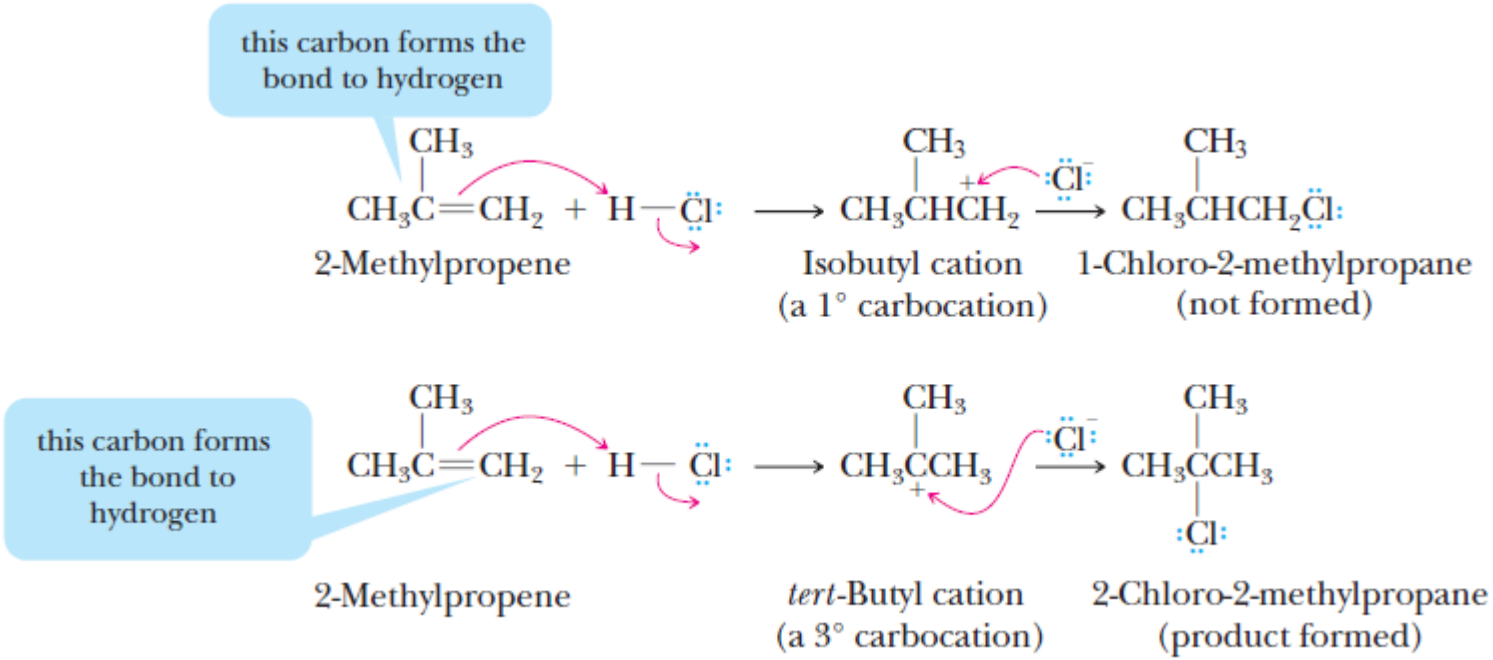
Regioselectivity and Markovnikov's rule

- The relative stabilities of carbocations is the underlying basis for Markovnikov's rule.
- The more stable carbocation is the one that is favoured;
- For example in the addition of HCl to propene:



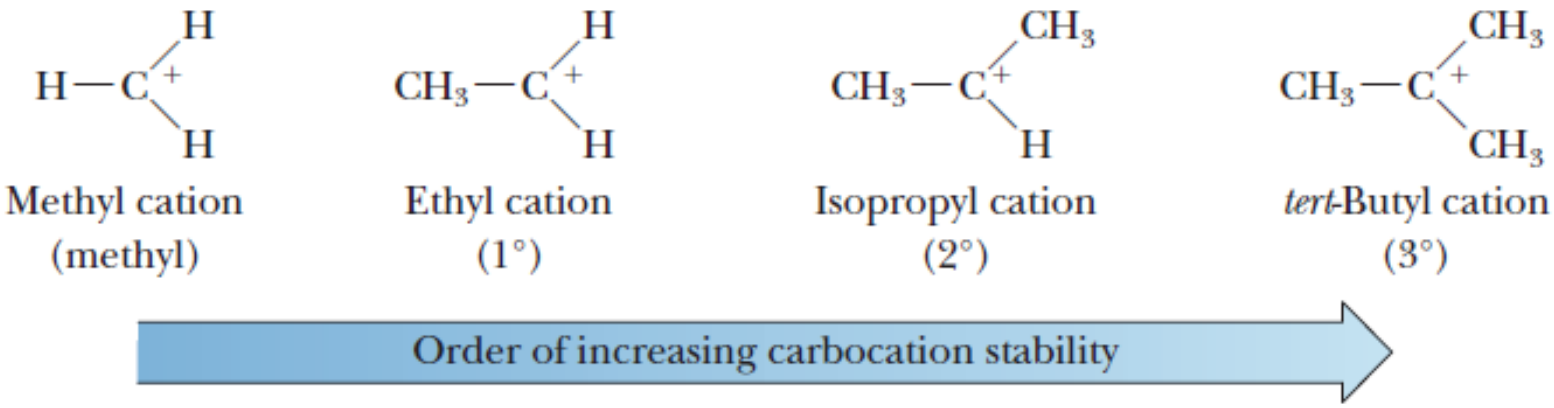
The 1° carbocation is less stable than the 2° carbocation. Therefore, 1-chloropropane is not formed.

- Similarly in the reaction of HCl with 2-methylpropene



The 3° carbocation is more stable than the 1° carbocation, therefore 2-chloro-2-methylpropane is formed.

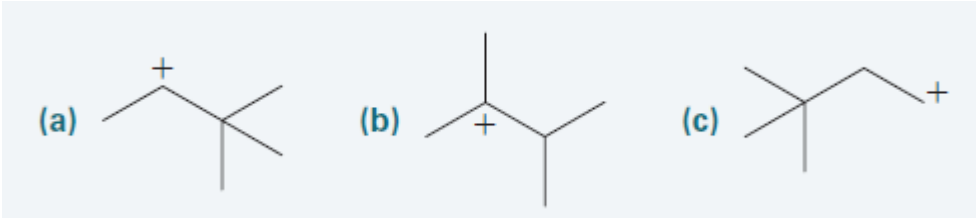
Recap from Lecture 1



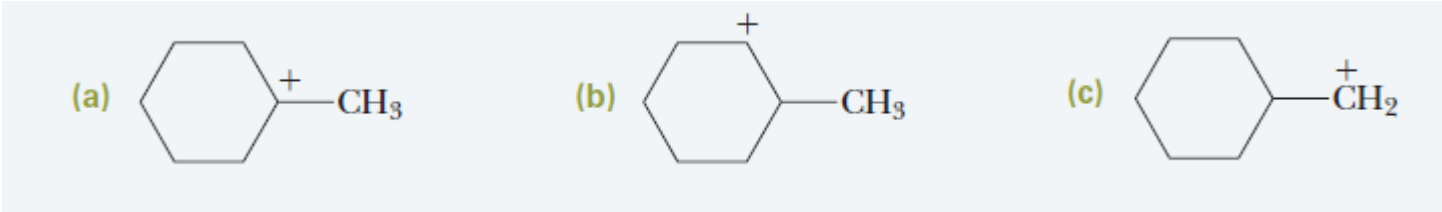
This happens due to the inductive effect. The positively charged carbon draws electron density towards itself from the alkyl groups (electron-donating groups). This allows the positive charge to be more delocalised – leading to stability. The more alkyl groups substituted, the greater the stability.

- Practice exercise 2: Arrange the following carbocations in order of increasing stability:

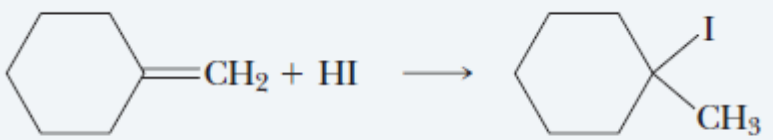
1)



2)



- Practice exercise 3: propose a mechanism for the addition of HI to methylenecyclohexane to give 1-iodo-1-methylcyclohexane:



Methylenecyclohexane 1-Iodo-1-methylcyclohexane

Free Radical Addition of HBr – Anti-Markovnikov Addition

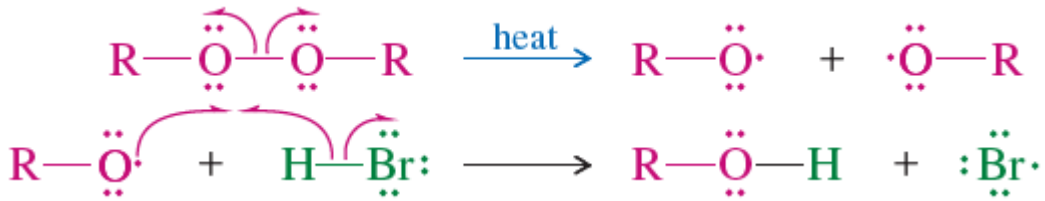
- In the presence of peroxides, the major product obtained from the addition of HBr to alkenes is the opposite of that expected from the Markovnikov's rule.

- This *Anti-Markovnikov addition* occurs by means of ***free radical addition mechanism***.

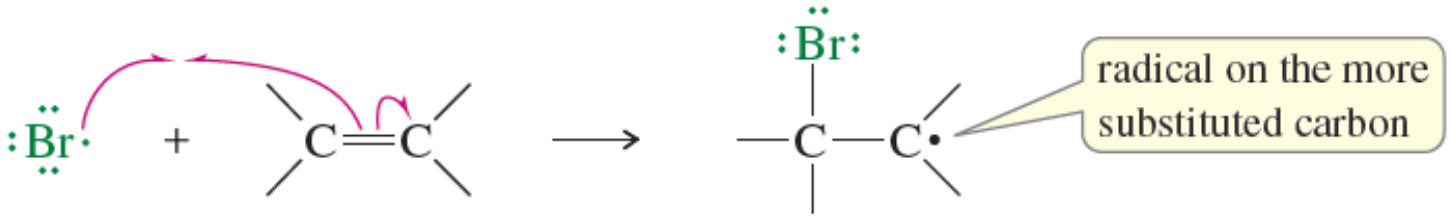
- The reaction is initiated by peroxides, which break down to give alkoxy radicals (R—O).

- The alkoxy radicals then cause the formation of bromine radicals.

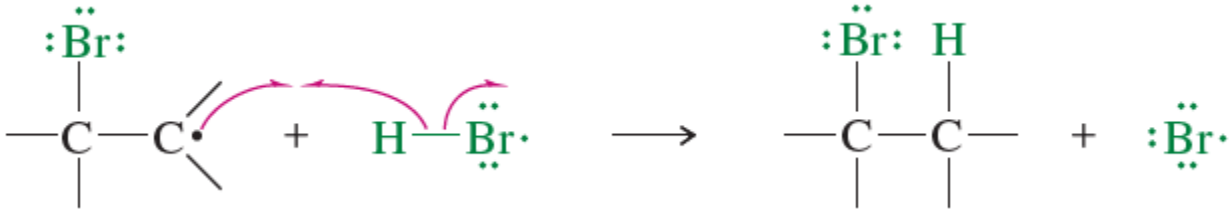
- *Step 1 Initiation*: Formation of radicals:



- *Step 2*: The bromine radical adds across the double bond to generate an alkyl radical on the more substituted carbon atom.



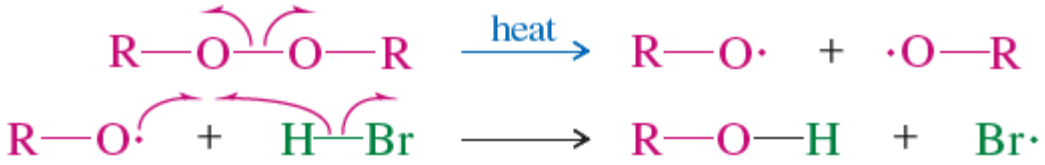
- *Step 3*: the alkyl radical abstracts a proton from H—Br to generate the product and a bromine radical.



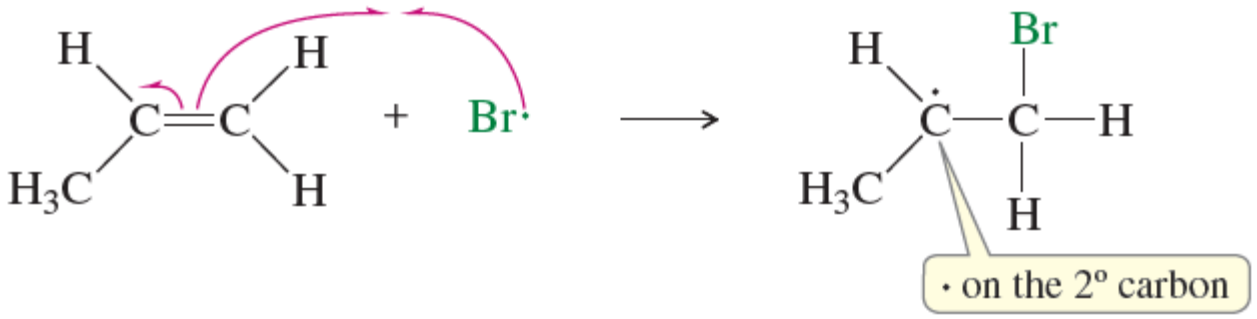
The bromine radical generated here reacts with another alkene molecule, continuing the chain.

- For example, free radical addition of HBr to propene in the presence of peroxide:

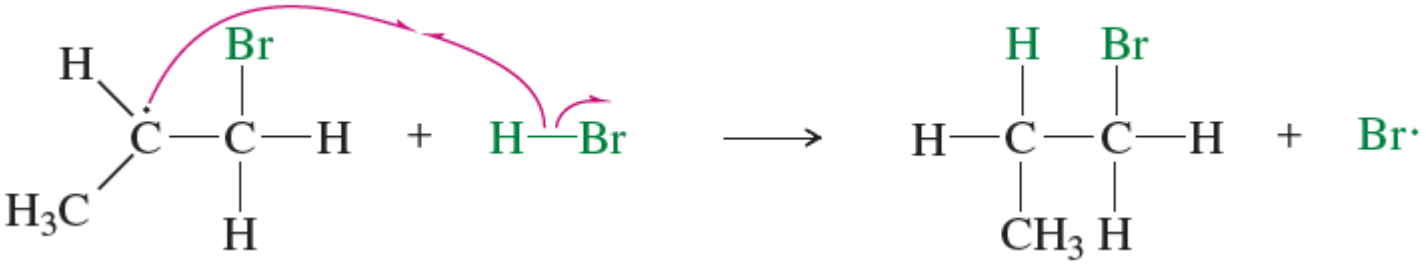
Initiation:



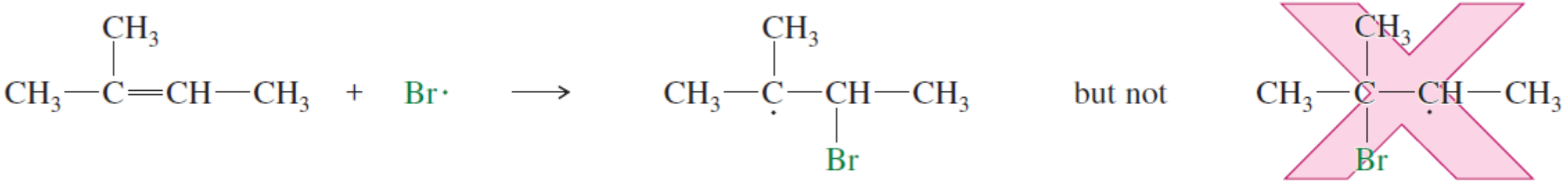
Step 1:



Step 2:



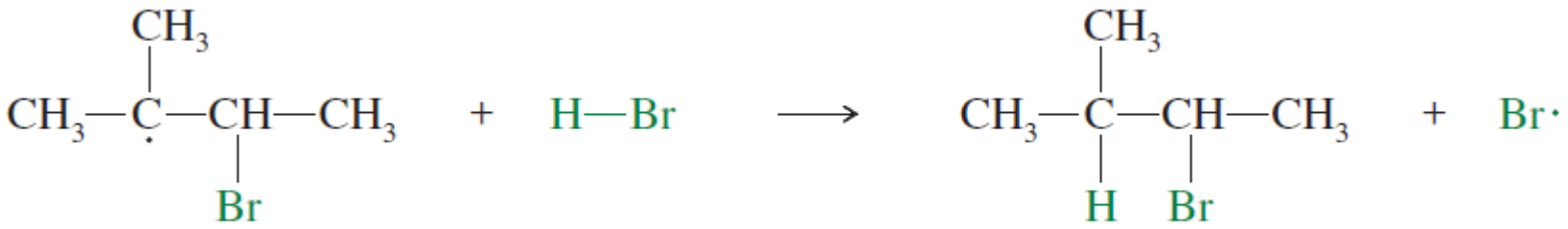
- With the unsymmetrical alkene 2-methylbut-2-ene, adding the bromine radical to the secondary end of the double bond forms the more stable tertiary radical.



tertiary radical (more stable)

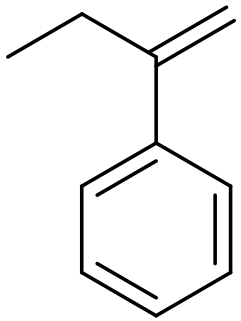
secondary radical (less stable)

- This radical intermediate then reacts with HBr to give the anti-Markovnikov product.



anti-Markovnikov product

- Practice exercise 4: but-1-ene-2-ylbenzene has the following structure:



a) Draw the displayed formula for the major product formed when but-1-ene-2-ylbenzene reacts with HBr in the presence of tert-butyl peroxide.

b) Draw out the mechanism of the reaction.

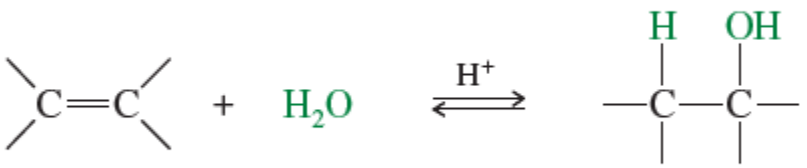
Addition of water to an alkene: Acid catalysed hydration

- In the presence of an acid catalyst, most commonly concentrated sulphuric acid (H_2SO_4), water adds to the C=C double bond to form an alcohol.

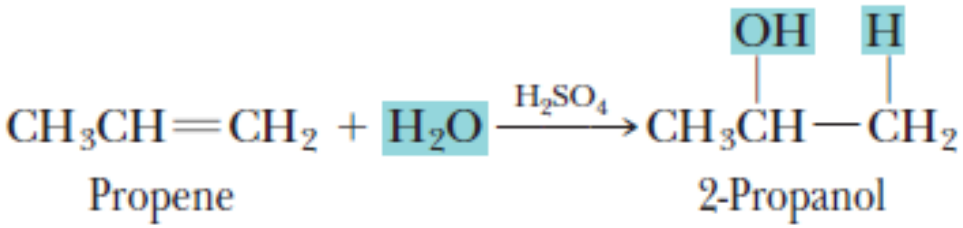
- This addition is known as a hydration, with a hydrogen atom adding to one carbon and a hydroxy group (OH) adding to the other carbon of the double bond.

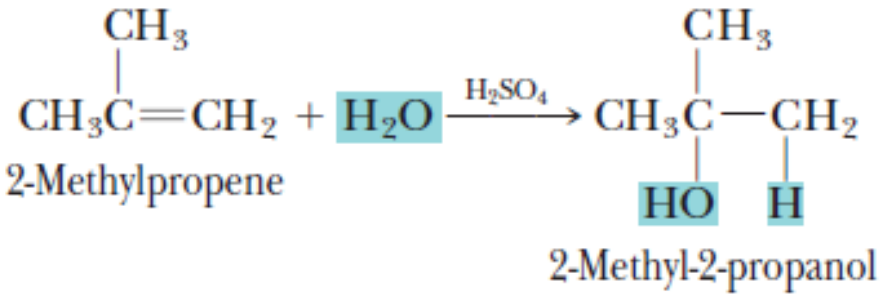
- Hydration of alkenes also follows Markovnikov's rule; the hydrogen adds to the carbon of the double bond that has the greater number of hydrogens.

Hydration of an alkene



(Markovnikov orientation)



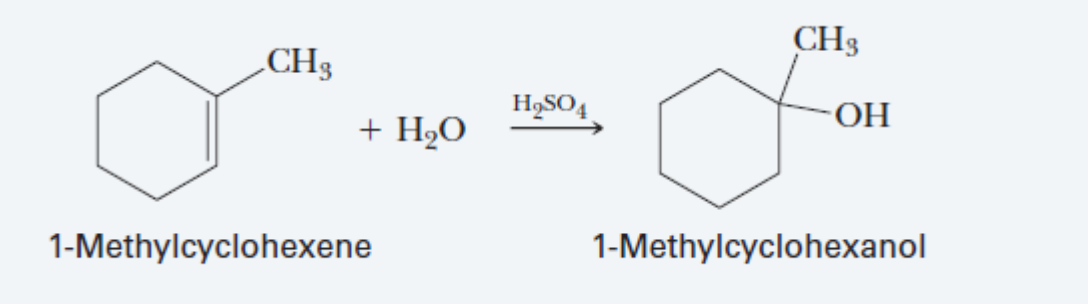


Hydration follows Markovnikov's rule.

Explained in terms of stabilities of carbocations.

- Practice exercise 5: Draw a skeletal formula for the product of the acid-catalysed hydration of 1-methylcyclohexene.

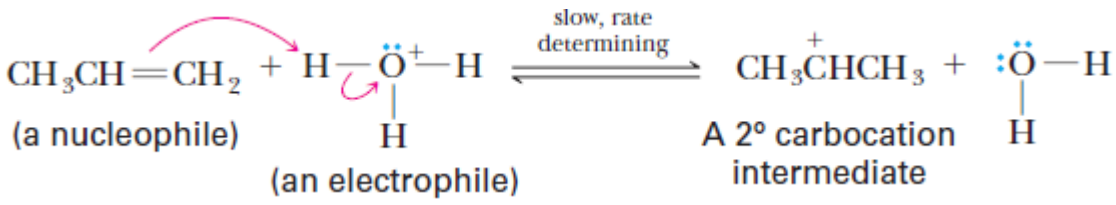
Ans:



Draw the skeletal formula for the product for each of the following alkene hydration reactions:

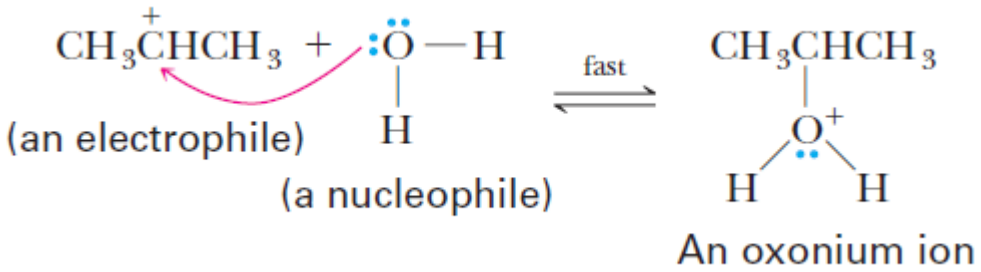
- The mechanism for the addition of H_2O to an alkene is similar to that of the addition of HX to an alkene.
- For example in the acid-catalysed hydration of propene:

Step 1: Add a proton: Note that in the presence of an acid catalyst, H_2O exists as the hydronium ion H_3O^+.



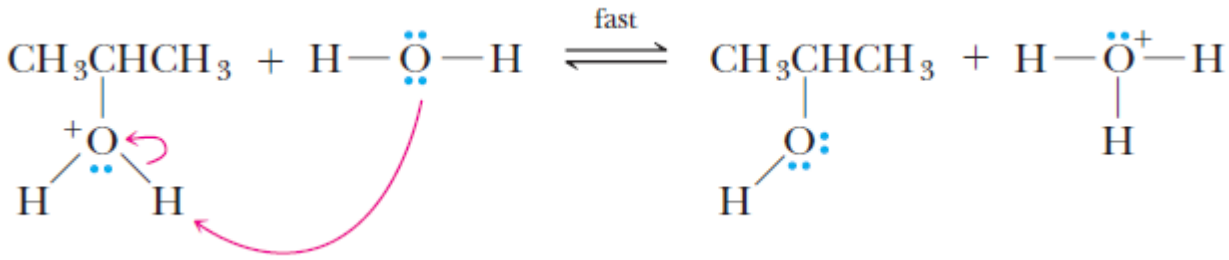
The electron pair in the double bond is used to form a new covalent bond with hydrogen from the hydronium ion. The result is a secondary carbocation (Markovnikov's rule)

Step 2: Reaction of a nucleophile and electrophile to form a new covalent bond.



The carbocation intermediate acts as an electrophile and the water that was generated in step 1 acts as a nucleophile. The lone pair of electrons on the oxygen of H_2O is used to form a new covalent bond with the positively charged carbon, to complete its valence shell. This results in the formation of an oxonium ion.

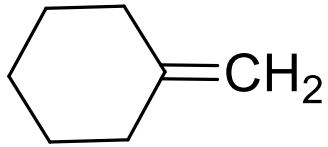
Step 3: Take a proton away:



Water in solution acts as a nucleophile and forms a covalent bond with the hydrogen from the oxonium ion. The electron pair in the O—H bond of the oxonium ion is transferred to the oxygen atom.

An alcohol is produced, and the acid catalyst is regenerated in the form of H_3O^+.

- Practice exercise 6: Propose a mechanism for the acid-catalysed hydration of methylenecyclohexane to give 1-methylcyclohexanol.



methylenecyclohexane